VOICEOPS | CAPSTONE REPORT

ZentroMart customer-support voice agent | Final verified snapshot | 25 October 2026

Project and four-step workflow

VoiceOps combines policy and catalog retrieval, deterministic rules, conversation state, and escalation tools. The stages below report saved evaluations and the verified synthetic speech run. No new LLM calls were made in this sprint.

Step	Method	Saved result
1 - Prompt-only	52 normal-text questions	23.1% content; 88.5% decision; 0.000 retrieval recall
2 - RAG	52 normal-text questions	96.2% content; 96.2% decision; 0.819 mean retrieval recall
3 - Conversations	96 turns across 23 conversations	99.0% checks/content; 100.0% decisions/tool calls; 22/23 fully correct
4 - Voice	3 synthetic turns, Gnani STT/TTS, offline rules agent	3 STT + 3 TTS calls; 0 errors; handoff ticket created

Saved Step 2 RAG improves over the Step 1 prompt-only baseline by 73.1 points in content accuracy and 7.7 points in decision accuracy. Step 1 and Step 2 are historical saved model-evaluation runs; the newer rules-only scores are separate.

Public repository: https://github.com/chukka-venugopalam/voiceops

Current quality and validation

Post-repair rules-only results are distinct from saved Step 1 and Step 2 baselines.

Regression and dataset checks

Check	Observed result
Full pytest	47 passed, 0 failed
Focused product/retrieval	23 passed
Dataset validator	PASS: 9 CSVs, 21 KB documents, 1,000 orders, 1,783 items, 244 logs, 52 questions, 23 conversations
Runner safeguards	Refuses existing output names; continues after file errors and exits nonzero; empty transcripts get clarification text/audio

Post-repair rules-only question evaluations

Field	Cases	Content	Decision	Mean recall
Normal text	52	100.0%	100.0%	0.989
ASR text	52	100.0%	100.0%	0.989

Conversations: 96 turns; 100.0% checks, 100.0% decisions, 100.0% content, 100.0% tool calls; 23/23 fully correct.

The repair aligned letter/digit catalog tokens, limited spoken "to" as a number to storage-capacity context, and added current electronics-return policy evidence for matched electronics. Historical-policy ranking remains date-sensitive; unspecified colors are not inferred.

Limitations: no new LLM calls in this sprint; browser microphone and personal recordings remain unverified. Saved RAG metrics are historical outputs, not a new live-model run.

Speech integration and reproducibility

Actual synthetic Gnani speech output | Shared session PC999

Three verified voice turns

Turn 1 - answer, 9.44s audio

Transcript: what is the return window for clothing
Assistant: I'm ZentroMart's AI support assistant. The return window starts on the delivery date and is counted in calendar days. The category of every product is listed in the `return_category` column of the product catalog.
Evidence IDs: RET-2, RET-9, RET-3, DWM-4

Turn 2 - answer, 14.08s audio

Transcript: does zentromart plus waive the return pickup fee
Assistant: Returns are collected from the delivery address through reverse pickup. A reverse pickup fee of INR 49 applies to change-of-mind returns of clothing, footwear and bags. The fee is waived for ZentroMart Plus members and for any item returned because it was damaged, defective or wrong.
Evidence IDs: RET-2, RET-6, PLS-2, FAQH-4

Turn 3 - escalate, 4.00s audio; ticket created

Transcript: please connect me to a human agent
Assistant: I'm sorry for the trouble. I've created a ticket and a human colleague will contact you.
Evidence IDs: FAQ-4, DWM-5, ESC-1, DWM-3

Install and run

Use Python 3.9 or newer. Extract the separate dataset (folders kb/, data/, practice/, tools/), create a virtual environment, install requirements.txt, copy blank .env.example to .env, and set DATASET_DIR and GNANI_API_KEY locally. Never commit credentials.

```
py -m venv .venv
.venv\Scripts\Activate.ps1
python -m pip install -r requirements.txt
Copy-Item .env.example .env
```

Audio-folder command (offline agent; Gnani speech services still required):

```
python run.py --audio-dir "samples\incoming" --output-dir "output" --language-code en-IN --offline
```

Each successful input produces reply text, WAV, and JSON with transcript, decision, evidence/chunk IDs and tool calls; results.csv and results.jsonl summarize the batch. Existing files are not overwritten. File errors are recorded; the batch continues and exits nonzero. Use --dry-run with same-name .txt files for testing without Gnani.

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The video uses actual synthesized replies from the synthetic turns. No answer was substituted. Results are not a guarantee of organizer scores.